

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
 May - June 2023
CV1001-1 – ELEMENTS OF CIVIL ENGINEERING



Duration: 3 Hours

Max. Marks: 100

Note:

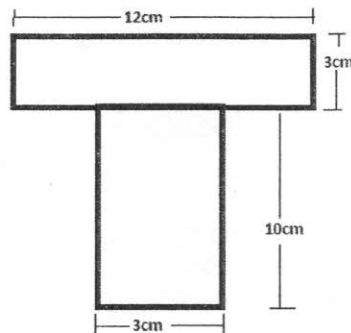
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

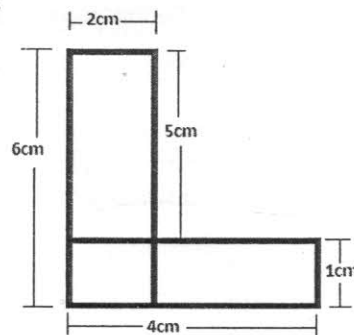
1. _____ is a branch of Civil Engineering dealing primarily with the design and construction of bridges, buildings and dams.
 A) Transportation Engineering B) Structural Engineering
 C) Geotechnical Engineering D) Construction Technology
2. _____ is a branch of Civil Engineering concerned with the conveyance of liquids through pipes and channels, especially as a source of mechanical force or control.
 A) Water Resources Engineering B) Environmental Engineering
 C) Hydraulics D) Construction Technology
3. _____ is a branch of Civil Engineering that involves the design of new systems and equipment that help manage human water resources.
 A) Construction Technology B) Hydraulics
 C) Environmental Engineering D) Water Resources Engineering
4. The primary objective of _____ is to create an optimal soil moisture regime for maximizing crop production and quality.
 A) Irrigation Engineering B) Hydraulics
 C) Geotechnical Engineering D) Environmental Engineering
5. _____ is applied while trying to turn a key into a lock.
 A) couple B) non-coplanar force
 C) momentum D) coplanar force
6. Couple is formed due to two _____.
 A) unlike, perpendicular and non-collinear forces of different magnitude B) like, parallel and non-collinear forces of same magnitude
 C) like, perpendicular and collinear forces of different magnitude D) unlike, parallel and non-collinear forces of same magnitude
7. Cycle pedaling is an example of _____.
 A) cyclic loading B) moment
 C) collinear forces D) couple
8. A couple produces _____.
 A) Motion combined translatory and rotational B) Rotational motion
 C) Translatory motion only D) Horizontal motion
9. Hinge support is called as _____.
 A) Ball joint B) Swivel joint
 C) Pin joint D) Socket joint
10. For a simply supported beam, moment at the support is always _____.
 A) Minimum B) Maximum
 C) Zero D) One
11. A hinged support doesn't offer resistance against _____.
 A) rotation B) horizontal movement
 C) vertical movement D) horizontal & vertical movement
12. Roller support is same as _____.
 A) Hinged Support B) Simply support
 C) Roller Support D) Fixed Support
13. The friction experienced by a body when it is in equilibrium is known as _____.
 A) Dynamic Friction B) Sliding Friction
 C) Static Friction D) Rolling Friction

14. The Friction experienced by a body when it is moving is called
 A) Sliding Friction B) Dynamic Friction
 C) Static Friction D) Rolling Friction
15. The dynamic friction experienced by a body when it slides over a plane is called
 A) Rolling Friction B) Static Friction
 C) Dynamic Friction D) Sliding Friction
16. The dynamic friction experienced by a body when it rolls over a surface
 A) Sliding Friction B) Static Friction
 C) Rolling Friction D) Dynamic Friction
17. Where will be the centroid of the T section lie from the base shown in the figure?



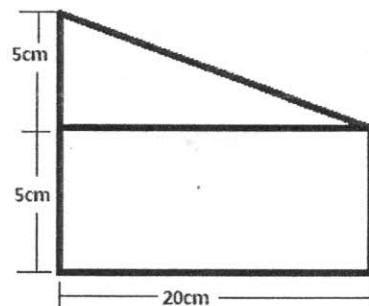
- A) At 8.545cm
 B) At 6.5 cm
 C) At 5 cm
 D) At 9.25 cm

18. Location of centroid of the L-section shown in the figure is



- A) (2.24 cm, 3.68 cm)
 B) (1.45 cm, 3.24 cm)
 C) (1.64 cm, 3.28 cm)
 D) (1.28 cm, 2.64 c

19. Where will be the centroid of the figure lies?



- A) (3.66cm, 8.84cm)
 B) (3.45cm, 4.52cm)
 C) (8.88 cm ,3.88 cm)
 D) (3.88cm, 8.88cm)

20. The use of centroid comes in picture as if the non-uniform loading is of the type of parabola then what will be the best suited answer among the following?
 A) The net force will act at the centroid of the parabola
 B) The net force will act the centre of the parabola
 C) The net force will act on the base of the parabola
 D) The net load will not be formed as all the forces will be cancelled

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the following scope of Civil Engineering i) Water resources engineering, ii) Surveying	4	L2	1	1,2
b)	Explain ideal conditions are assumed while applying the principles of mechanics to solve the practical problems (basic idealization).	4	L2	1	1,2
c)	Find the reaction at the contact points of two identical cylinders of weight 100N each placed in a trench as shown in Fig. Q1(c).	8	L3	2	1,2

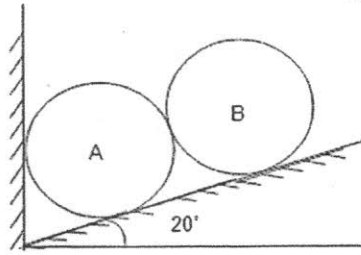


Fig. Q1(c)

2. a) Define free body diagram. Explain with any two examples. 4 L2 2 1,2
 b) What is Couple? What are its characteristics? 4 L2 2 1,2
 c) Determine the magnitude of unknown force and resultant force in a system of force as shown in Fig. Q2(c) whose resultant is a horizontal force.

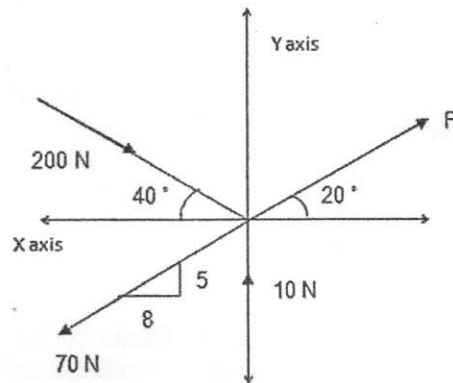


Fig. Q2(c)

3. a) Define force. Mention the characteristics of force with neat sketch. 4 L2 1 1,2
 b) Explain any four force system with neat sketches and examples. 4 L2 1 1,2
 c) Sketch the resultant of force system acting on the plate element with respect to point A as shown in Fig. Q3(c).
 Given the grid size 100mm x 100 mm.

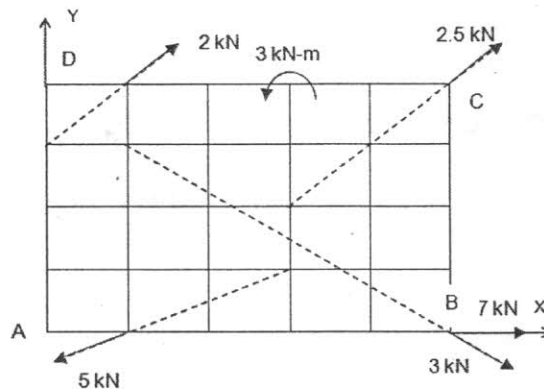


Fig. Q3(c)

Unit - II

4. a) State Coulomb's laws of friction (any four). 4 L2 4 1,2
 b) Define i) Angle of repose and ii) angle of friction. 4 L1 4 1,2
 c) Determine the supports reactions for the loaded beam shown in Fig. Q4(c).

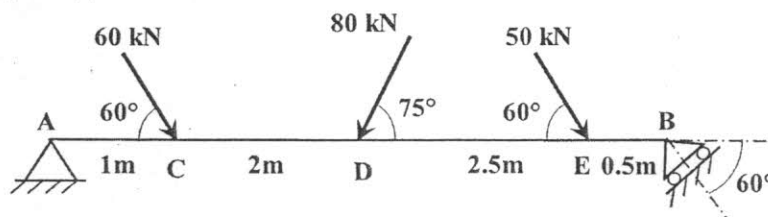


Fig. Q4(c)

8 L3 1 1,2

4 L2 1 1,2

4 L2 1 1,2

8 L3 2 1,2

4 L2 4 1,2

4 L1 4 1,2

8 L3 3 1,2

5. a) Explain cantilever and simply supported beams with examples.
 b) Explain Uniformly distributed and Uniformly Varying load with neat sketches.
 c) Two blocks A & B weighing 4kN & 2.5kN, respectively, are connected by a wire passing over a smooth frictionless pulley as shown in the Fig. Q5(C). Determine the magnitude of force P which will cause block B to impend towards right. Take $\mu=0.2$ for all the contact surfaces.

4	L2	3	1,2
4	L2	3	1,2

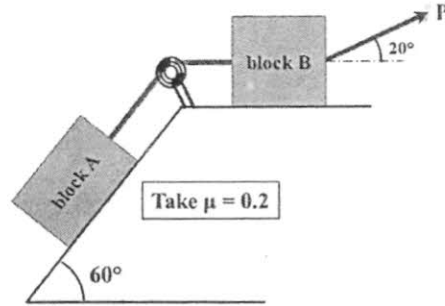


Fig. Q5(c)

6. a) Define statically determinate block/beam with an example.
 b) Explain the different types of supports with the neat sketches (any four).
 c) A ladder AB of length 5m weighing 250N is placed against a vertical wall and a horizontal ground. A man of 750N is standing at the top of the ladder. Determine the minimum angle made by ladder to the ground so that ladder does not slip. The coefficient of friction between ladder & the wall is 0.2 and that between ladder & floor is 0.3.

8	L3	4	1,2
4	L2	3	1,2
4	L2	3	1,2

Unit - III

7. a) State and prove parallel axis theorem.
 b) Derive an expression for moment of inertia of a triangle about the base by method of integration.
 c) Locate the centroid of the shaded area about the axes shown in the Fig. Q7(c) below.
8. a) Distinguish between centroid and center of gravity.
 b) Derive an expression for centroid of a Quarter circle radius 'r' from the first principles.
 c) Determine the moment of inertia about the axis AB for the plane as shown in Fig. Q8(c) below.

8	L3	4	1,2
4	L2	5	1,2
4	L3	5	1,2
8	L3	5	1,2
4	L2	5	1,2
4	L3	5	1,2
8	L3	5	1,2

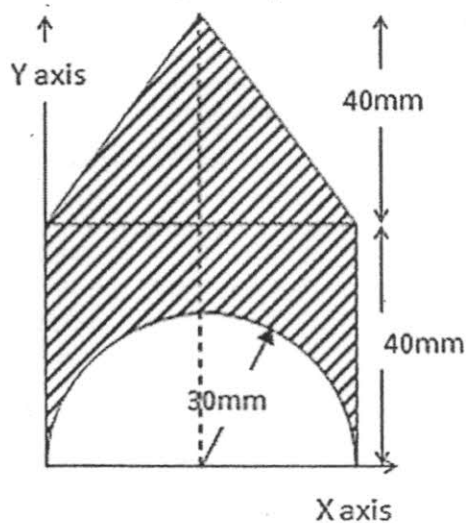


Fig. Q7(c)

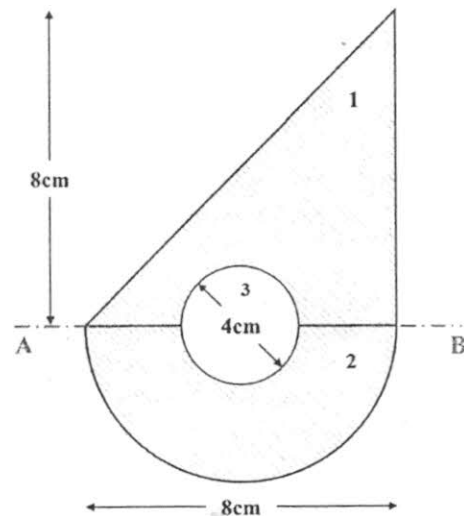


Fig. Q8(c)

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
